

Voice-Driven Multi-Agent Test & Debugging Swarm

IBM Bob 2.0 Hackathon | September 2026

HridaySharma2002 | github.com/HridaySharma2002/voice-driven-test-swarm

The Problem: Context-Switching Tax

When a CI/CD build fails, developers must manually:

1. Parse RestAssured / Selenium execution logs
2. Trace the failure back through Spring Boot logic
3. Cross-reference schemas, configs, and test reports
4. Write and apply a fix, then re-run tests

This multi-step debugging loop inflates Mean Time to Resolution (MTTR) and drains developer focus from building features.

The Solution: One Voice Command, Full Resolution

The Voice-Driven Swarm collapses the entire debugging loop into a single spoken trigger:

```
"Run API regression tests."
```

The system then autonomously:

- Executes the hybrid Java test suite (RestAssured + Selenium)
- Parses Maven Surefire XML failure reports
- Hands the stack trace to IBM Bob 2.0 for root-cause analysis
- Plays the fix summary out loud via ElevenLabs TTS

Result: a multi-hour debugging cycle reduced to under 60 seconds.

Architecture — 4 Phases

Phase	Component	Technology	Role
1	Target App	Java 21, Spring Boot 3.2, Maven	Buggy REST API + disabled UI button causes intentional failures + Surefire XML
2	Voice Orchestrator	Python	Recognises spoken command, triggers pipeline (mic simulated on Python 3.14)
3	Connector	Python + xml.etree	Runs mvn clean test, parses failure elements from surefire XML reports
4	Bob Handoff + TTS	IBM Bob 2.0 + ElevenLabs v2	Bob analyses repo, explains root cause; Sarah voice plays summary aloud

The Deliberate Bug — CheckoutController.java

The controller uses `flat subtraction` instead of percentage-based discount math:

```
// DELIBERATE BUG
double finalPrice = order.getPrice() - order.getDiscount();
```

```
if (finalPrice != 90.0) return ResponseEntity.badRequest();
```

Test sends price=200, discount=10:

BUG: $200 - 10 = 190$ which is not 90.0 so returns HTTP 400 Test FAILS

FIX: $200 \times (1 - 10/100) = 180$ returns HTTP 200 Test PASSES

IBM Bob 2.0 Usage — Evaluator + Optimizer Pattern

In Bob IDE (23.93 Bobcoins, 12 tasks completed):

- Fixed ElevenLabs v1 to v2 SDK migration
- Resolved pyaudio build failure on Python 3.14 (no MSVC wheel available)
- Fixed relative cwd path errors using `os.path.abspath(__file__)`
- Renamed CheckoutAPITest.java to CheckoutApiTest.java (Java filename rule)
- Fixed pom.xml missing spring-boot-starter-test dependency
- Hardcoded Maven absolute path (not on system PATH)

In the pipeline (`bob_handoff.py`):

- bob run called with the full Surefire XML failure trace as a natural language prompt
- Bob reasons over the Spring Boot repository and returns a 2-sentence fix summary
- Fallback feeds directly into ElevenLabs TTS when CLI path is restricted

Live Terminal Output

```
[MIC BYPASSED] Simulating voice command capture...
```

```
Recognized: 'run api regression tests'
```

```
Intent recognized: Triggering CI/CD Pipeline...
```

```
Executing Java hybrid test suite...
```

```
Detected 2 test failures. Handing off to IBM Bob 2.0...
```

```
[IBM Bob 2.0 Summary]: The test failed because
```

```
CheckoutController uses flat subtraction instead of
```

```
percentage-based discount math. Fix: price*(1-discount/100)
```

```
Playing audio summary...
```

Impact & Submission Details

Before: 30-60 min manual debug cycle

After: Under 60 seconds, fully autonomous, voice-activated

GitHub: github.com/HridaySharma2002/voice-driven-test-swarm

Bob Evidence: `bob_sessions/` — 2 task screenshots, 23.93 Bobcoins total

Stack: Java 21 | Spring Boot 3.2 | Maven | RestAssured | Selenium 4

Python 3.14 | IBM Bob 2.0 | ElevenLabs v2 SDK (Sarah voice)